

FEA Implementation Details for TWR_and_MaxPayload.m Program

1.1 Constants & Parameters Setup

- Defines base drone specifications: frame/electronics mass (1.0 kg), estimated base cost (\$50.00), and quadcopter arms (4).
- Establishes physical constants ($g = 9.81 \text{ m/s}^2$) and operational criteria: thrust per motor (9.81 N) and target Thrust-to-Weight Ratio ($\text{TWR} = 2.0$).

1.2 Geometry & Volume Calculation

- Reads arm geometry from STL files (CircleArm.STL / TriangleArm.STL).
- Computes exact volume using the Signed Tetrahedron Volume Method:

$$V = \frac{1}{6} \sum |\vec{v}_1 \cdot (\vec{v}_2 \times \vec{v}_3)|$$

- Standardizes unit conversions between mm^3 and m^3 .

1.3 CSV Material Data Parsing

- Dynamically imports Materials.csv mapping:
 - Material Name, Density (kg/m^3), Cost per Unit Mass (USD/kg)
 - Young's Modulus (E), Poisson's Ratio (ν), and Yield Strength (σ_y)

1.4 Mesh Generation & Boundary Conditions Setup

- Initializes a 3D static solid model using MATLAB's PDE Toolbox (createpde).
- Generates a tetrahedral mesh with max element size $H_{\max} = 0.002$ m
- Visualizes geometry face IDs (pdegplot) and mesh topology (pdeplot3D).
- Calculates motor surface area (Face 10) to apply surface traction pressure:

$$\text{Pressure} = \frac{\text{Thrust}}{\text{Area}}$$

- Fixes body mounting bolt faces (Face 5).

1.5 Multi-Material Loop & FEA Solver

1. Updates elasticity properties (E and ν) per row in the database.
2. Solves static equilibrium: $\mathbf{Ku} = \mathbf{F}$.
3. Extracts von Mises stress (σ_{vm}) and displacement magnitude (u_{mag}).
4. Evaluates Factor of Safety ($\text{FoS} = \sigma_{\text{yield}}/\sigma_{\text{vm,max}}$).
5. Reports payload capacity and total system cost.